



Data Base
CCE 333

Database Management System for a Sports Club

Third Year Computer and Control Engineering

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1. Introduction

1.1 Project Description

This project presents the complete design and implementation of a Database Management System for a Sports Club. The system is designed to manage members, coaches, staff, sports activities, facilities, equipment, payments, and member dependents. The database captures all essential relationships between these entities, enabling efficient data storage, retrieval, and management for the club's daily operations.

1.2 Problem Statement

Managing a sports club involves handling a large volume of interconnected data: member registrations, enrollment in various sports, coach assignments, facility bookings, equipment tracking, and payment processing. Without a structured database system, this data becomes prone to redundancy, inconsistency, and loss. This project addresses these challenges by designing a normalized relational database that ensures data integrity, eliminates redundancy, and supports efficient querying for all club operations.

2. Conceptual Design

2.1 Entity-Relationship Diagram (ERD)

The following ERD represents the conceptual design of the Sports Club database using Chen Notation. It identifies all entities, their attributes (including derived, multivalued, and composite attributes), and the relationships between them with their cardinality and participation constraints.

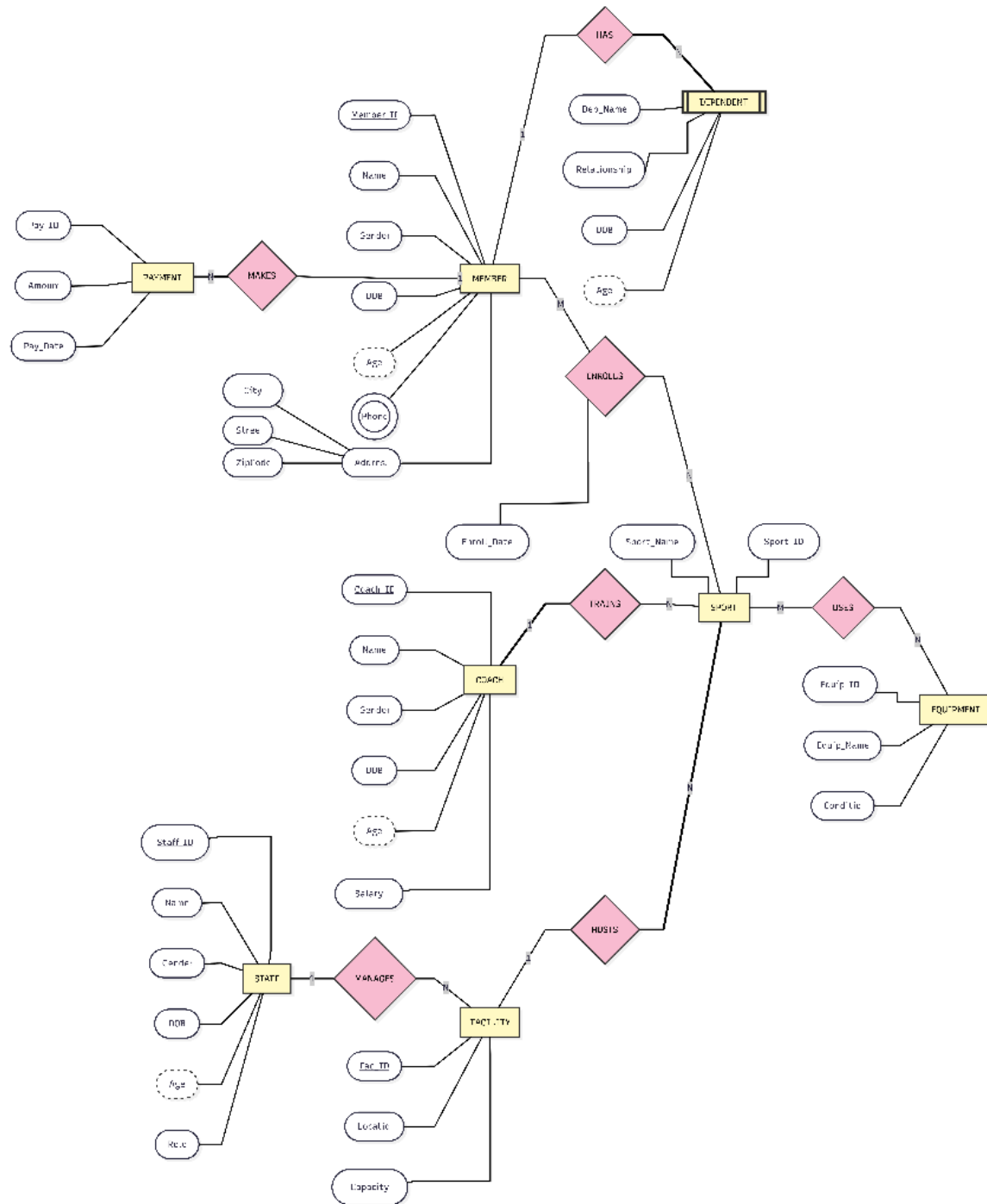


Figure 2.1: Entity-Relationship Diagram (Chen Notation)

ERD Entities & Relationships Summary

Entity	Type	Primary Key	Key Attributes
MEMBER	Strong	Member_ID	Name, Gender, DOB,

DEPENDENT	Weak	(Member_ID, Dep_Name)	Phone(MV), Address(Composite)
PAYMENT	Strong	Pay_ID	Amount, Pay_Date
SPORT	Strong	Sport_ID	Sport_Name
COACH	Strong	Coach_ID	Name, Gender, DOB, Salary
EQUIPMENT	Strong	Equip_ID	Equip_Name, Condition
FACILITY	Strong	Fac_ID	Location, Capacity
STAFF	Strong	Staff_ID	Name, Gender, DOB, Role

Relationship	Between	Type	Participation
HAS	Member - Dependent	1:N (Identifying)	Total for Dependent
MAKES	Member - Payment	1:N	Partial for Member
ENROLLS	Member - Sport	M:N	Partial for both
TRAINS	Coach - Sport	1:N	Total for Coach
USES	Sport - Equipment	M:N	Partial for both
HOSTS	Facility - Sport	1:N	Total for Sport
MANAGES	Staff - Facility	1:N	Partial for both

2.2 Enhanced Entity-Relationship Diagram (EERD)

The EERD introduces a generalization hierarchy. MEMBER, COACH, and STAFF share common attributes (Name, Gender, DOB, Age) which are abstracted into a PERSON superclass. The specialization is Total (every person must belong to at least one subclass) and Overlapping (a person can be both a Member and a Coach, for example).

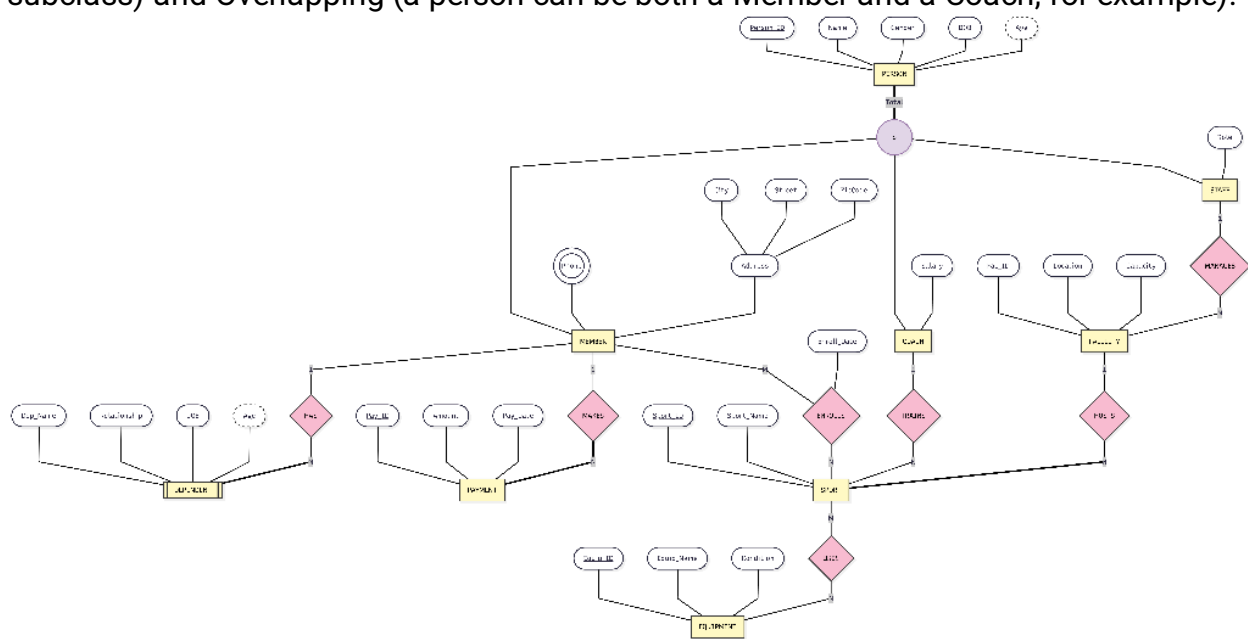


Figure 2.2: Enhanced Entity-Relationship Diagram (EERD)

Constraint	Value	Meaning
Completeness	Total	Every Person MUST belong to at least one subclass
Disjointness	Overlapping (o)	A Person CAN belong to multiple subclasses simultaneously

3. Logical Design

3.1 Relational Schema (Mapping Rules)

The following mapping rules were applied to convert the ERD to a relational schema:

- Each strong entity maps to its own table with its primary key.
- Weak entity maps to a table with composite PK (owner PK + partial key).
- 1:N relationships: FK placed on the N-side table.
- M:N relationships: new junction (associative) table with composite PK.
- Multivalued attributes: separate table with composite PK.
- Composite attributes: decomposed into atomic columns.
- Derived attributes: NOT stored (computed at query time).

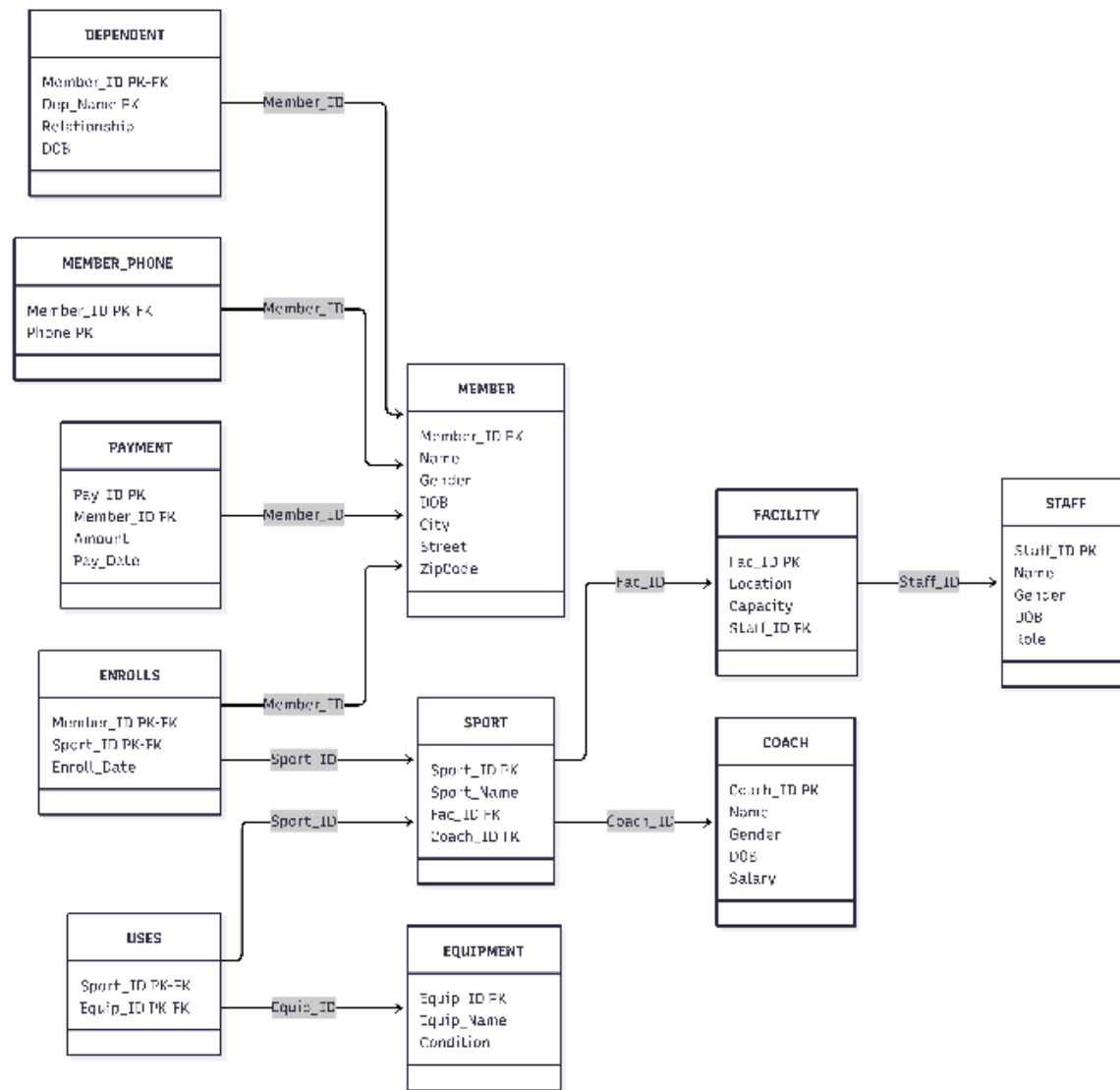


Figure 3.1: Relational Schema Diagram

Schema Definition

#	Table Name	Columns
1	MEMBER	Member_ID (PK), Name, Gender, DOB, City, Street, ZipCode
2	MEMBER_PHONE	Member_ID (PK,FK), Phone (PK)
3	DEPENDENT	Member_ID (PK,FK), Dep_Name (PK), Relationship, DOB
4	PAYMENT	Pay_ID (PK), Member_ID (FK), Amount, Pay_Date
5	ENROLLS	Member_ID (PK,FK), Sport_ID (PK,FK), Enroll_Date
6	SPORT	Sport_ID (PK), Sport_Name, Fac_ID (FK), Coach_ID (FK)
7	COACH	Coach_ID (PK), Name, Gender, DOB, Salary
8	EQUIPMENT	Equip_ID (PK), Equip_Name, Condition
9	USES	Sport_ID (PK,FK), Equip_ID (PK,FK)
10	FACILITY	Fac_ID (PK), Location, Capacity, Staff_ID (FK)
11	STAFF	Staff_ID (PK), Name, Gender, DOB, Role

3.2 Database Normalization

3.2.1 First Normal Form (1NF)

Rule: Every attribute must contain only atomic (indivisible) values. No repeating groups.

Table	1NF	Justification
MEMBER	PASS	Phone (multivalued) extracted to MEMBER_PHONE. Address decomposed to City, Street, ZipCode. Age (derived) not stored.
MEMBER_PHONE	PASS	Each row stores one phone number. Composite PK (Member_ID, Phone).
DEPENDENT	PASS	All attributes atomic. Age (derived) not stored.
PAYMENT	PASS	All attributes atomic.
ENROLLS	PASS	All attributes atomic.
SPORT	PASS	All attributes atomic.
COACH	PASS	All attributes atomic. Age (derived) not stored.
EQUIPMENT	PASS	All attributes atomic.
USES	PASS	All-key relation, no non-key attributes.
FACILITY	PASS	All attributes atomic.
STAFF	PASS	All attributes atomic. Age (derived) not stored.

3.2.2 Second Normal Form (2NF)

Rule: Must be in 1NF AND every non-key attribute must be fully functionally dependent on the entire primary key (no partial dependencies). Only relevant for composite PKs.

• Single-PK tables (MEMBER, PAYMENT, SPORT, COACH, EQUIPMENT, FACILITY, STAFF): Partial dependency is structurally impossible. 2NF satisfied trivially.

Table (Composite PK)	Analysis	2NF
MEMBER_PHONE (Member_ID, Phone)	No non-key attributes exist.	PASS
DEPENDENT (Member_ID, Dep_Name)	Relationship and DOB depend on full composite key, not on any single part.	PASS
ENROLLS (Member_ID, Sport_ID)	Enroll_Date depends on full composite key (which member enrolled in which sport).	PASS
USES (Sport_ID, Equip_ID)	No non-key attributes exist.	PASS

3.2.3 Third Normal Form (3NF)

Rule: Must be in 2NF AND no non-key attribute may depend on another non-key attribute (no transitive dependencies).

Table	Functional Dependencies	Transitive Dep?	3NF
MEMBER	Member_ID -> Name, Gender, DOB, City, Street, ZipCode	None found	PASS
PAYMENT	Pay_ID -> Member_ID, Amount, Pay_Date	None	PASS
SPORT	Sport_ID -> Sport_Name, Fac_ID, Coach_ID	FKs don't determine other non-key attrs	PASS
COACH	Coach_ID -> Name, Gender, DOB, Salary	None	PASS
EQUIPMENT	Equip_ID -> Equip_Name, Condition	None	PASS
FACILITY	Fac_ID -> Location, Capacity, Staff_ID	None	PASS
STAFF	Staff_ID -> Name, Gender, DOB, Role	None	PASS
DEPENDENT	(Member_ID, Dep_Name) -> Relationship, DOB	None	PASS
ENROLLS	(Member_ID, Sport_ID) -> Enroll_Date	None	PASS

Conclusion: All 11 tables satisfy 1NF, 2NF, and 3NF.

4. Physical Implementation (SQL)

4.1 Data Definition Language (DDL)

The following SQL statements create all database tables with proper constraints:

```
-- =====  
-- Sports Club Database - DDL (CREATE TABLES)  
-- =====
```

```
CREATE DATABASE IF NOT EXISTS sports_club;  
USE sports_club;
```

-- 1. MEMBER

```
CREATE TABLE MEMBER (  
    Member_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Name VARCHAR(100) NOT NULL,  
    Gender CHAR(1) CHECK (Gender IN ('M','F')),  
    DOB DATE NOT NULL,  
    City VARCHAR(50),  
    Street VARCHAR(100),  
    ZipCode VARCHAR(10)  
);
```

-- 2. MEMBER_PHONE

```
CREATE TABLE MEMBER_PHONE (  
    Member_ID INT,  
    Phone VARCHAR(20),  
    PRIMARY KEY (Member_ID, Phone),  
    FOREIGN KEY (Member_ID) REFERENCES MEMBER(Member_ID) ON DELETE CASCADE  
);
```

-- 3. DEPENDENT

```
CREATE TABLE DEPENDENT (  
    Member_ID INT,  
    Dep_Name VARCHAR(100),  
    Relationship VARCHAR(50) NOT NULL,  
    DOB DATE,  
    PRIMARY KEY (Member_ID, Dep_Name),  
    FOREIGN KEY (Member_ID) REFERENCES MEMBER(Member_ID) ON DELETE CASCADE  
);
```

-- 4. STAFF

```
CREATE TABLE STAFF (  
    Staff_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Name VARCHAR(100) NOT NULL,  
    Gender CHAR(1) CHECK (Gender IN ('M','F')),  
    DOB DATE,  
    Role VARCHAR(50)
```

);

– 5. FACILITY

```
CREATE TABLE FACILITY (  
    Fac_ID    INT PRIMARY KEY AUTO_INCREMENT,  
    Location  VARCHAR(100) NOT NULL,  
    Capacity  INT,  
    Staff_ID  INT,  
    FOREIGN KEY (Staff_ID) REFERENCES STAFF(Staff_ID) ON DELETE SET NULL  
);
```

– 6. COACH

```
CREATE TABLE COACH (  
    Coach_ID  INT PRIMARY KEY AUTO_INCREMENT,  
    Name      VARCHAR(100) NOT NULL,  
    Gender    CHAR(1) CHECK (Gender IN ('M','F')),  
    DOB       DATE,  
    Salary    DECIMAL(10,2)  
);
```

– 7. SPORT

```
CREATE TABLE SPORT (  
    Sport_ID  INT PRIMARY KEY AUTO_INCREMENT,  
    Sport_Name VARCHAR(100) NOT NULL,  
    Fac_ID    INT,  
    Coach_ID  INT,  
    FOREIGN KEY (Fac_ID) REFERENCES FACILITY(Fac_ID) ON DELETE SET NULL,  
    FOREIGN KEY (Coach_ID) REFERENCES COACH(Coach_ID) ON DELETE SET NULL  
);
```

– 8. ENROLLS (Junction: Member M:N Sport)

```
CREATE TABLE ENROLLS (  
    Member_ID INT,  
    Sport_ID  INT,  
    Enroll_Date DATE NOT NULL,  
    PRIMARY KEY (Member_ID, Sport_ID),  
    FOREIGN KEY (Member_ID) REFERENCES MEMBER(Member_ID) ON DELETE CASCADE,  
    FOREIGN KEY (Sport_ID) REFERENCES SPORT(Sport_ID) ON DELETE CASCADE  
);
```

– 9. PAYMENT

```
CREATE TABLE PAYMENT (  
    Pay_ID    INT PRIMARY KEY AUTO_INCREMENT,  
    Member_ID INT NOT NULL,  
    Amount    DECIMAL(10,2) NOT NULL,  
    Pay_Date  DATE NOT NULL,  
    FOREIGN KEY (Member_ID) REFERENCES MEMBER(Member_ID) ON DELETE CASCADE  
);
```

– 10. EQUIPMENT

```
CREATE TABLE EQUIPMENT (  
    Equip_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Equip_Name VARCHAR(100) NOT NULL,  
    Condition_ VARCHAR(50)  
);
```

– 11. USES (Junction: Sport M:N Equipment)

```
CREATE TABLE USES (  
    Sport_ID INT,  
    Equip_ID INT,  
    PRIMARY KEY (Sport_ID, Equip_ID),  
    FOREIGN KEY (Sport_ID) REFERENCES SPORT(Sport_ID) ON DELETE CASCADE,  
    FOREIGN KEY (Equip_ID) REFERENCES EQUIPMENT(Equip_ID) ON DELETE CASCADE  
);
```

4.2 Data Manipulation Language (DML)

The following INSERT statements populate the database with sample data:

– Sports Club Database - DML (INSERT DATA)

– MEMBER

INSERT INTO MEMBER (Member_ID, Name, Gender, DOB, City, Street, ZipCode) VALUES

(1, 'Ahmed Hassan', 'M', '1998-03-15', 'Cairo', 'Nasr City St.', '11765'),
(2, 'Sara Mohamed', 'F', '2000-07-22', 'Giza', 'Dokki St.', '12311'),
(3, 'Omar Ali', 'M', '1999-11-05', 'Cairo', 'Heliopolis St.', '11341'),
(4, 'Nour Ibrahim', 'F', '2001-01-18', 'Alexandria', 'Smouha St.', '21500'),
(5, 'Youssef Khaled', 'M', '1997-09-30', 'Cairo', 'Maadi St.', '11728');

– MEMBER_PHONE

INSERT INTO MEMBER_PHONE (Member_ID, Phone) VALUES

(1, '01012345678'), (1, '01112345678'),
(2, '01023456789'), (3, '01034567890'),
(4, '01045678901'), (4, '01145678901'),
(5, '01056789012');

– DEPENDENT

INSERT INTO DEPENDENT (Member_ID, Dep_Name, Relationship, DOB) VALUES

(1, 'Layla Ahmed', 'Daughter', '2020-05-10'),
(1, 'Fatma Hassan', 'Wife', '1999-08-20'),
(3, 'Ali Omar', 'Son', '2022-01-15'),
(5, 'Mona Khaled', 'Wife', '1998-12-01');

– STAFF

INSERT INTO STAFF (Staff_ID, Name, Gender, DOB, Role) VALUES

(1, 'Mahmoud Samir', 'M', '1985-06-12', 'Manager'),
(2, 'Heba Farouk', 'F', '1990-03-25', 'Receptionist'),
(3, 'Tamer Adel', 'M', '1988-11-08', 'Maintenance');

– FACILITY

INSERT INTO FACILITY (Fac_ID, Location, Capacity, Staff_ID) VALUES

(1, 'Main Building - Hall A', 200, 1),
(2, 'Swimming Pool Complex', 50, 2),
(3, 'Outdoor Football Field', 300, 3),
(4, 'Tennis Court Area', 30, 1);

– COACH

INSERT INTO COACH (Coach_ID, Name, Gender, DOB, Salary) VALUES

(1, 'Karim Mostafa', 'M', '1980-04-20', 5000.00),
(2, 'Dina Nabil', 'F', '1985-09-14', 4500.00),
(3, 'Hassan Magdy', 'M', '1978-12-01', 6000.00),
(4, 'Rania Saeed', 'F', '1990-07-30', 4000.00);

– SPORT

```
INSERT INTO SPORT (Sport_ID, Sport_Name, Fac_ID, Coach_ID) VALUES
(1, 'Football', 3, 1),
(2, 'Swimming', 2, 2),
(3, 'Tennis', 4, 3),
(4, 'Basketball', 1, 4);
```

– ENROLLS

```
INSERT INTO ENROLLS (Member_ID, Sport_ID, Enroll_Date) VALUES
(1, 1, '2025-01-10'), (1, 2, '2025-02-15'),
(2, 2, '2025-01-20'), (2, 3, '2025-03-01'),
(3, 1, '2025-01-12'), (4, 4, '2025-02-28'),
(5, 3, '2025-01-05'), (5, 1, '2025-03-10');
```

– PAYMENT

```
INSERT INTO PAYMENT (Pay_ID, Member_ID, Amount, Pay_Date) VALUES
(1, 1, 500.00, '2025-01-10'),
(2, 1, 300.00, '2025-02-15'),
(3, 2, 500.00, '2025-01-20'),
(4, 3, 500.00, '2025-01-12'),
(5, 4, 400.00, '2025-02-28'),
(6, 5, 600.00, '2025-01-05');
```

– EQUIPMENT

```
INSERT INTO EQUIPMENT (Equip_ID, Equip_Name, Condition_) VALUES
(1, 'Football', 'Good'),
(2, 'Swimming Goggles', 'New'),
(3, 'Tennis Racket', 'Good'),
(4, 'Basketball', 'Used'),
(5, 'Goal Net', 'Good');
```

– USES

```
INSERT INTO USES (Sport_ID, Equip_ID) VALUES
(1, 1), (1, 5),
(2, 2),
(3, 3),
(4, 4);
```

4.3 Database Execution Screenshots (phpMyAdmin)

[INSERT YOUR phpMyAdmin SCREENSHOTS HERE]

Instructions: After running the DDL and DML queries in phpMyAdmin:

1. Take a screenshot showing all created tables in the database
2. Take screenshots showing the data inside each table
3. Replace this placeholder page with those screenshots

[PAGE 2 - MORE SCREENSHOTS]

5. Software Implementation (GUI)

5.1 Application Interface Overview

A simple web-based graphical user interface (GUI) was developed to allow data entry into the Sports Club database. The interface provides forms for inserting new members, enrollments, payments, and other records directly into the database.

[INSERT YOUR GUI SCREENSHOTS HERE]

5.2 Front-end to Database Integration

The web interface connects to the MySQL database using PHP (mysqli). Each form submits data via POST requests, which are processed by PHP scripts that execute the corresponding INSERT queries on the database.

[INSERT INTEGRATION SCREENSHOTS HERE]

6. Conclusion

This project successfully demonstrated the complete lifecycle of database design and implementation for a Sports Club management system. Starting from the conceptual design phase with an Entity-Relationship Diagram (ERD) using Chen Notation, we identified eight core entities – Member, Dependent, Payment, Sport, Coach, Equipment, Facility, and Staff – along with their attributes and inter-relationships.

The Enhanced ERD (EERD) introduced a generalization hierarchy by abstracting shared attributes into a PERSON superclass with Total, Overlapping specialization constraints, reflecting the real-world scenario where a person can simultaneously hold multiple roles (e.g., a coach who is also a club member).

The logical design phase mapped the conceptual model to a relational schema consisting of 11 normalized tables. All tables were verified to satisfy First, Second, and Third Normal Forms (1NF, 2NF, 3NF), ensuring the elimination of data redundancy and the preservation of data integrity.

The physical implementation translated the schema into executable SQL code (DDL for table creation, DML for data population), which was successfully deployed on a MySQL database through phpMyAdmin. A simple web-based GUI was also developed to facilitate data entry operations.

This project reinforced key database engineering concepts including normalization theory, ER modeling, schema mapping rules, and the practical bridge between theoretical design and working software systems.